

GPU	Operator	Evaluated Shapes	Lines of Code			Performance Baseline	Normalized Performance	
			CUDA	Triton	Hexcute		Triton	Hexcute
NVIDIA A100 GPU	FP16 GEMM	40	703 ^b	71	98	cuBLAS	0.75×	1.01 ×
	Fused MHA ^a Forward	20	577	114	172	FlashAttention2	0.91×	1.49 ×
	Fused MHA Decoding	24	322	224	253	FlashInfer	0.50×	1.02 ×
NVIDIA H100 GPU	Blockwise Scaled FP8 GEMM	35	900	87	180	CUTLASS	0.50×	1.17 ×
	Warp Specialized FP16 GEMM	40	1024 ^b	71	169	cuBLAS	0.61×	1.27 ×
	Fused MHA Forward	20	1684	114	212	FlashAttention3	0.54×	1.21 ×

^a Multi-head Attention

^b For GEMM, speedups are reported against cuBLAS, while LoC comparisons use CUTLASS because cuBLAS is closed-source.